Standard Operating Procedure — Backup Verification

Standard Operating Procedure: Nightly Database Backup Verification

Company: Halcyon Logistics

Document Purpose: This SOP outlines the procedures for verifying nightly database backups to ensure data integrity, recovery readiness, and compliance with organizational standards.

Purpose

Nightly database backups are critical to safeguarding Halcyon Logistics' operational data. This procedure ensures backups are complete, consistent, and recoverable, minimizing the risk of data loss and ensuring business continuity. Verification is required to confirm backups meet quality standards and are stored securely.

Scope

This SOP applies to all database systems managed by Halcyon Logistics, including but not limited to the Order Management System (OMS), Inventory Tracking Database (ITD), and Customer Relationship Management (CRM) platform. It applies to all IT staff, database administrators (DBAs), and backup administrators responsible for backup processes.

Roles and Responsibilities

- **Database Administrators (DBAs):** Execute backup verification tasks, document results, and escalate issues.
- **Backup Administrator:** Ensure backup tools are configured correctly and accessible for verification.
- **IT Support Team:** Provide technical assistance during verification and escalate unresolved issues.
- **IT Manager:** Approve changes to verification procedures and oversee compliance.

Prerequisites

Before initiating verification:

1. 1. Confirm the backup schedule aligns with the nightly backup plan.
2. 2. Ensure all backup tools (e.g., SQL Server Backup, AWS S3, or Azure Blob Storage) are operational.
3. 3. Verify access to backup storage locations and recovery tools.
4. 4. Review the latest backup logs for errors or warnings.

Procedure

5. 1. **Initiate Verification:** At 2:00 AM daily, the backup administrator triggers the verification process via the automated backup tool.
6. 2. **Check Backup Logs:** Review the backup log files for errors, warnings, or incomplete backups. Log any discrepancies.
7. 3. **Verify File Integrity:** Use checksum tools (e.g., MD5 or SHA-256) to compare backup file hashes against pre-defined values.
8. 4. **Test Restoration:** Select a sample database (e.g., a non-production environment) and attempt to restore the backup. Confirm the restored database matches the source data.
9. 5. **Document Results:** Record verification outcomes in the IT Operations Dashboard, including success status, any issues, and corrective actions.
10. 6. **Notify Stakeholders:** If verification is successful, notify the IT Manager. If issues are found, escalate per the Escalation section.

Verification Checks

- **Backup Completeness:** Ensure all tables, indexes, and stored procedures are included.
- **Data Consistency:** Validate that restored data matches the source database's current state.
- **Storage Compliance:** Confirm backups are stored in approved locations with proper retention policies.
- **Performance Metrics:** Check backup size, duration, and resource usage against baseline thresholds.

Escalation

- **Minor Issues (e.g., minor corruption):** Resolve within 2 hours; document in the IT ticketing system.
- **Major Issues (e.g., failed restoration):** Escalate to the IT Manager within 1 hour. If unresolved, notify the CTO and initiate a root-cause analysis.
- **Critical Failures (e.g., storage outage):** Escalate to the IT Manager immediately; activate the disaster recovery plan.

Revision History

- **Version 1.0 (2023-10-05):** Initial release.
- **Version 1.1 (2024-02-15):** Updated verification checks and escalation procedures.
- **Version 1.2 (2024-08-20):** Added compliance with GDPR data retention policies.

This SOP is reviewed annually or upon significant system changes to ensure alignment with Halcyon Logistics' operational standards.